

Geological, Geotectonic & Seismic Setting

Assessment of Earthquake & Tsunami Threat and Identification of Safe Zones for Floating Production Unit (FPU) Deployment — Southern Coast of Africa (Namibia)

Region	Offshore Namibia — Southern Atlantic Passive Margin
Basins Assessed	Orange, Luderitz, Walvis, Namibe
Scope	Geology Geotectonics Seismicity Tsunami Hazard FPU Zone ID
Hazard Framework	PSHA (2023), PSHA 10% exceedance in 50 years
Key Standards	International Seismological Centre (ISC), UNESCO-IOC, NOAA

1. REGIONAL GEOLOGICAL FRAMEWORK

1.1 Basement Geology and Tectonic Domains

Namibia's basement geology is dominated by the Kalahari Craton and surrounding mobile belts. The country encompasses four major tectonic domains, each with distinct seismic behaviour and relevance to offshore siting decisions:

Tectonic Domain	Location	Age	Seismic Relevance
Congo Craton	Northern Namibia	Archaean	Low hazard; stable nucleus
Kalahari Craton	Southern Namibia	~1800 Ma	Stable — offshore S. coast sits here
Kaoko Belt	NW margin	Pan-African	Moderate activity; not offshore
Damara Belt	Central Namibia	Pan-African	Mobile belt — moderate hazard inland
Namaqua-Natal Belt	SW Namibia	Mesoproterozoic	NW-SE gneissic foliation; Hebron Fault

The southern coastal zone where FPU deployment is planned lies over the southern portion of the Kalahari Craton — the most tectonically stable basement unit in Namibia. The passive Atlantic continental margin offshore was formed during the Early Cretaceous breakup of Gondwana and the opening of the South Atlantic, creating a thick post-rift sedimentary succession that now hosts significant hydrocarbon resources.

1.2 Passive Margin Architecture

The Orange Basin, the primary focus for FPU deployment, forms part of the Southwest African Coastal Basin lying offshore Namibia and western South Africa, extending from the Walvis Ridge in the north to the Agulhas Falkland Fracture Zone in the south. Key characteristics:

- Total offshore basin coverage: ~825,418 km² (0–4,000 m water depth)
- Drift succession: thick seaward-thinning sedimentary wedge overlying rift basins, with depocentre defined by the Kudu Basement Arch (S. Namibia) and Agulhas Columbine Arch to the south

- Stratigraphic sequence: Lower Cretaceous (Barremian) aeolian sandstones and Aptian–Albian turbidite systems — providing stable, well-consolidated seabed substrate at FPU mooring depths
- No active subduction or convergent plate boundary offshore Namibia; the Mid-Atlantic Ridge (a spreading centre) is the nearest active plate boundary, thousands of km distant

2. GEOTECTONIC SETTING

2.1 Regional Stress Field

Southern Africa's neotectonic environment is governed by two competing far-field stress systems that drive the region's intraplate seismicity:

Stress System	Orientation	Type	Effect in Namibia
East African Rift System (EARS)	NNE extensional	Normal faulting	Extends geophysically into Namibia; drives graben formation
Wegener Stress Anomaly (WSA)	NW-WNW compressive	Strike-slip	Anomalous for a passive margin; principal driver of Namibia seismicity

The WSA is particularly relevant because it is anomalous: a passive margin should exhibit extensional stress, yet a NW-directed compressive anomaly exists across western Namibia and the adjacent southeast Atlantic offshore. This creates strike-slip fault reactivation potential along older inherited structures, particularly the NW-SE Namaqua foliation in the southwest.

2.2 Key Structural Features

- **Hebron / Dreylingen Fault System:** A 45 km-long neotectonic scarp in SW Namibia, NW-SE trending, with three left-stepping segments. Identified via Landsat imagery and high-resolution DEM. Exceptionally preserved due to hyper-arid Namib Desert conditions and slow erosion rates (0.1–3.6 m/Myr). Capable of large-magnitude intraplate events with very long recurrence intervals.
- **Kamanjab Inlier Fault Cluster:** A Paleoproterozoic crystalline basement block in northern Namibia with increasing seismicity since 2018. NE-dipping seismogenic fault produced the largest instrumentally recorded Namibia earthquake (Mw 5.4, April 2021).
- **Great Escarpment:** Major topographic feature separating the elevated interior plateau from the coast. Escarpment retreat (Cretaceous) and associated stress unloading may contribute to localized fault reactivation in the coastal hinterland.

3. SEISMIC HAZARD ASSESSMENT

3.1 Probabilistic Seismic Hazard (PSHA) Results

A comprehensive country-wide PSHA for Namibia (2023) used earthquake catalogue data spanning 1910–2021, delineating 11 seismic source zones plus the Hebron/Dreylingen fault source. Four Ground Motion Prediction Equations (GMPEs) were applied with equal weighting using a logic-tree formalism. Key outputs for 10% probability of exceedance in 50 years (475-year return period):

Zone / Region	PGA Range (g)	Hazard Level	FPU Relevance
Orange Basin offshore (S. coast)	0.017 – 0.045	LOW	PRIMARY ZONE — optimal

Luderitz Basin offshore	0.030 – 0.055	LOW	SECONDARY ZONE — suitable
SW inland (Hebron Fault area)	0.060 – 0.090	MEDIUM	Onshore buffer zone — avoid within 50 km
Central Namibia	0.070 – 0.100	MEDIUM-HIGH	Inland only — not relevant to offshore
NW Namibia (Kamanjab)	0.100 – 0.149	HIGH	Northern coast — avoid
North-central Namibia	0.085 – 0.130	HIGH	Inland — not relevant to offshore

3.2 Historical and Instrumental Seismicity

The earthquake record for Namibia (1910–2021) is characterised by low to moderate intraplate events. Notable events:

- Mw 5.6, 31 July 2009 — largest historical event
- Mw 5.4, 4 April 2021 — Kamanjab Inlier; largest instrumentally recorded; produced ~9,000 detectable aftershocks and related events in a 3-month window
- Mw 5.6, 4 April 2021 (second event same day) — confirmed active fault cluster in northern basement inlier
- Southern and offshore zones have remained seismically quiet throughout the instrumental record, consistent with Kalahari Craton stability

Focal mechanisms across southern Africa, including Namibia, predominantly indicate normal faulting — consistent with EARS-influenced extensional tectonics. The offshore passive margin has no documented seismogenic fault systems.

4. TSUNAMI HAZARD ASSESSMENT

4.1 South-East Atlantic Context

The southeast Atlantic Ocean, where Namibia's coast is located, is one of the world's lowest-tsunami-risk ocean margins. Unlike the Pacific with its subduction-dominated Ring of Fire, the Atlantic is bounded by passive margins and a spreading Mid-Atlantic Ridge, precluding the mega-thrust earthquakes that generate catastrophic tsunamis. Comparative probabilistic data:

Ocean Basin / Region	Damaging Tsunami Probability (50 yr)	Primary Source Types
Pacific (outer coasts)	40 – 80%	Subduction mega-thrusts, volcanic collapse
Atlantic (outer coasts)	1 – 15%	Submarine landslides, distal EQs
SE Atlantic / Namibia coast	<5% (estimated)	Remote sources only
UNESCO-IOC: Namibia national	>10% (medium classification)	Conservative national estimate

4.2 Credible Tsunami Sources Affecting Namibia

Source	Distance	Max Estimated Runup	Warning Time	Risk to FPU
Canary Islands flank collapse (La Palma scenario)	~5,000 km	Uncertain; locally large	~6–8 hours	LOW — debated; long warning time
Bay of Cadiz / Azores earthquake zone	~6,500 km	1–3 m at coast	~8–10 hours	VERY LOW — attenuated over distance
Scotia Arc / S. Sandwich Is. subduction	~4,000 km	<1 m estimated	~4–6 hours	VERY LOW — infrequent, southerly
SW Namibia intraplate fault (Hebron/Dreylingen)	Inland / coastal	Uncertain; historical only	Minutes (local)	LOW offshore — fault is onshore
Mid-Atlantic Ridge spreading events	>3,000 km	Negligible	N/A	NEGLIGIBLE — spreading, not thrust

A critical operational advantage: all credible remote tsunami sources are at least 4,000 km from the Namibian coast, providing 4–10 hours of warning time. This is sufficient for an FPU to offload personnel, close safety systems, and reposition to deep water if required under emergency protocols. The UNESCO-IOC Tsunami Warning System covers the South Atlantic through the Pacific Tsunami Warning Centre (PTWC).

5. OFFSHORE SEDIMENTARY BASINS

Namibia has four offshore sedimentary basins covering approximately 825,418 km², all on the passive Atlantic margin, all tectonically stable:

Basin	Location	Water Depth	Key Feature	FPU Status
Orange Basin	S. Namibia – N. South Africa	0 – 4,000 m	Venus (TotalEnergies); Giant light oil	FPSO planned; FID 2026
Luderitz Basin	Central S. coast	0 – 3,000 m	Kudu gas field; Barremian sandstones	FPU ~170 km offshore planned
Walvis Basin	Central coast	0 – 3,500 m	Mopane; Wingat-1 oil confirmed	Exploration/appraisal phase
Namibe Basin	N. Namibia	0 – 3,000 m	Highly underexplored	Pre-exploration

6. RECOMMENDED ZONES FOR SAFE FPU DEPLOYMENT

ZONE 1 — Orange Basin Ultradeepwater

Location: ~290–320 km SW of Luderitz, Block 2913B | **Water depth:** 2,600–3,300 m

Seismic hazard: PGA <0.030 g — lowest in Namibia. No active offshore faults. Stable Cretaceous turbidite substrate. Distance from all mapped neotectonic scarps exceeds 200 km.

Tsunami hazard: Very low for deepwater location; all sources provide >4 hours warning. FPU can be repositioned to survivable heading.

Precedent: TotalEnergies Venus FPSO development (up to 40 subsea wells, 160,000 bbl/d) advancing to FID 2026 confirms engineering feasibility and regulatory pathway at this exact location.

Verdict: PRIMARY RECOMMENDATION — optimal combination of seismic safety, tsunami resilience, proven substrate conditions, and commercial activity.

ZONE 2 — Luderitz Basin / Kudu Field Area

Location: ~170 km offshore of Luderitz (Elizabeth Bay area) | **Water depth:** ~200–500 m (shallow-water setting)

Seismic hazard: PGA 0.030–0.055 g — low. Overlies stable Barremian sandstone reservoir sequence. Shallower water provides better mooring certainty and easier geotechnical investigation.

Tsunami hazard: Moderate coastal exposure (medium UNESCO-IOC classification) but adequate warning time from all remote sources. Shallower water may amplify wave height relative to deepwater Zone 1 — site-specific inundation modelling recommended.

Precedent: BW Energy floating production platform planned at this location for Kudu gas-to-power project; pipeline to Elizabeth Bay under development planning.

Verdict: SECONDARY RECOMMENDATION — suitable for gas/production FPU; slightly higher coastal tsunami exposure than Zone 1 warrants detailed site assessment.

ZONES TO AVOID OR TREAT WITH CAUTION

- **NW Kamanjab coast region:** Highest seismic hazard in Namibia (PGA up to 0.149 g); active fault cluster; increasing seismicity trend since 2018. Unsuitable for any fixed or floating offshore infrastructure.
- **Within 50 km of Hebron / Dreylingen fault scarps (SW Namibia inland):** Neotectonic evidence of large-magnitude intraplate earthquakes with extremely long recurrence intervals. Low probability but high consequence; detailed paleoseismic assessment required before any onshore/near-shore infrastructure.
- **Great Escarpment coastal hinterland:** Stress unloading from ongoing geomorphic processes may reactivate inherited structural features. Geotechnical investigation recommended for any near-shore pipeline landing or onshore facility.

7. CONCLUSIONS

Namibia presents a geologically favourable environment for offshore Floating Production Unit deployment on the southern African Atlantic coast. The key findings are summarised below:

Finding	Assessment	Implication for FPU
Plate tectonic setting	Stable Atlantic passive margin; no active subduction offshore	No mega-thrust earthquake risk; no tsunami generation offshore
Basement geology	Kalahari Craton underlies S. coast; oldest, most stable basement in Namibia	Minimal ground amplification; predictable foundation conditions
Seismic hazard (offshore S.)	PGA 0.017–0.045 g (10% / 50 yr); lowest in Namibia	FPU structural design requirements are minimal vs. Pacific equivalents

Key inland faults	Hebron/Dreylingen (SW) and Kamanjab cluster (N); not present offshore	FPU located >200 km from nearest active fault — negligible risk transfer
Tsunami hazard	Medium national classification; very low for deepwater offshore sites	4–10 hours warning from all credible sources; FPU repositioning feasible
Preferred deployment zone	Orange Basin (~320 km SW Luderitz, 2,600–3,300 m depth)	Zone 1 — primary recommendation (Venus FPSO precedent)
Secondary deployment zone	Luderitz Basin (~170 km offshore, ~200–500 m depth)	Zone 2 — suitable with site-specific tsunami inundation modelling

References & Key Sources

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- NAMCOR Offshore Basin History. Orange, Luderitz, Walvis, Namibe basins; 825,418 km² coverage; Kudu 9A-1 discovery 1973.
- BW Energy / NAMCOR Kudu Field Development. FPU ~170 km offshore; Elizabeth Bay pipeline; 420 MW Phase 1 gas-to-power.